

Explainable machine learning analysis of factors associated with skilled birth attendance in Burkina Faso

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Abstract

Skilled birth attendance (SBA) is essential for reducing maternal and neonatal mortality, but its coverage varies in Burkina Faso. This study used explainable machine learning to predict SBA, identify key predictors, and examine geographic and urban–rural descriptive inequalities. This cross-sectional study analyzed 5,111 women aged 15–49 years from the 2021 Burkina Faso Demographic and Health Survey (BF-DHS). Features were selected using Boruta feature selection, and SMOTE used for class imbalance. Five machine learning models used in this study such as Random Forest (RF), Decision Tree (DT), K-Nearest Neighbors (KNN), Logistic Regression (LR), and Support Vector Machine (SVM) were trained. Model performance was evaluated using accuracy, precision, recall, F1-score, Matthew’s correlation coefficient (MCC), Cohen’s kappa, and AUROC. SHapley Additive exPlanations (SHAP) interpreted model predictions, while decision curve analysis (DCA) and spatial mapping evaluated clinical utility and geographic variation. Among five models, RF achieved the highest discrimination, with moderate performance (AUROC=0.71). The key predictors included province, antenatal care visits ≥ 4 , maternal age at first birth ≥ 20 years, age at first sexual intercourse ≥ 18 years, sexual activity, household wealth, and religion. Descriptive spatial mapping showed the higher predicted SBA probabilities in central and western provinces and lower probabilities in Sahel, Sud-Ouest, and Est, with greater inequalities in rural areas. These results highlight that interpretable machine learning combined with spatial analysis can identify predictors of SBA and capture subnational inequalities, offering reproducible, data-driven insights for maternal health research.

Keywords SBA · Burkina faso · Machine learning · SHAP · Spatial

Introduction

Maternal healthcare utilization, particularly SBA, remain discrepant across low- and middle-income countries (LMICs), with substantial urban–rural and provincial disparities¹. SBA defined as delivery care provided by trained and accredited health professionals such as midwives, nurses, or physicians is widely recognized as a critical intervention for reducing maternal and neonatal mortality². Despite global efforts under the Millennium Development Goals (MDGs) and Sustainable Development Goals (SDGs), preventable maternal deaths continue

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to pose a major public health challenge, particularly in sub-Saharan Africa (SSA), where progress has been inconsistent and coverage of SBA remains limited³. Maternal mortality is a very infrequent occurrence and hard to quantify accurately particularly in the areas where there are weak civil registration systems⁴. As a result, the rate of births that are attended to by skilled staff associates with the rate of improvement on maternal health⁵. SBA needs a helpful environment besides the availability of a competent provider such as necessary drugs, operational referral systems, emergency obstetric care (EDO), and conducive health policies⁶. The empirical data have always indicated the existence of an inverse association between SBA coverage and maternal mortality, making it the core element of maternal health policies⁷. In SSA, several studies used machine methods to maternal healthcare utilization, wash components, and related outcome, including prediction of home delivery and health access behaviors⁸. Additionally, machine learning used widely in DHS-based public health research have revealed the efficacy of predictive modeling and spatial inference in LMICS⁹. However, the combination of explainable machine learning methods, imbalance-aware modeling methods, and spatial prediction mapping for SBA in Burkina Faso remains limited¹⁰.

Burkina Faso is a low-income nation in West Africa that is still facing significant maternal and neonatal mortality rates despite the gradual improving in the coverage of maternal health services over the last ten years¹¹. The coverage of SBA by the nation has improved but there still exist significant disparities that cut across geographic location, socioeconomic status and place of residing¹². The rural women, lower educated; lower household wealth, higher parity, and poor access to antenatal care continue to be disproportionately associated with SBA¹³. These differences are further associated by cultural practices, early marriages, lack of independence in decision making, distance to health facilities, and the lack of money¹⁴. It is necessary to understand the role of these associated factors in determining the use of SBA in order to provide and evidence based maternal and health policy¹⁵. Although prior studies have identified associates of SBA using traditional regression approaches, few have leveraged interpretable machine learning to uncover the nuanced behavioral predictors associated maternal healthcare decisions at the population level, and even fewer have integrated spatial and subgroup analyses to contextualize these patterns¹⁶. To address this gap, this study used five machine learning models such as RF, DT, KNN, LR, and SVM on nationally DHS data to identify key predictors of SBA using SHAP interpretability analyses¹⁷. These methods provide to quantify both global and local contributions of socio-demographic and reproductive factors, revealing behavioral predictors such as maternal education, age at first birth, antenatal care utilization, and socio-economic status¹⁸. Furthermore, province-level predictive mapping and urban–rural subgroup analyses highlighted heterogeneity in maternal healthcare decision-making, providing spatially and contextually nuanced insights into behavioral patterns and inequities in SBA utilization¹⁹. Recent developments in machine learning (ML) provide strong options that can overcome these constraints. RF, and SVM algorithms have the benefit of being flexible in modelling nonlinear relationships and can automatically learn to capture interactions and improve predictive accuracy with no rigid distributional assumptions²⁰. Combined with interpretable ML approaches like SHAP, which identify both global and local predictor contributions, these techniques can be used to measure the determinants of behavior in maternal healthcare decisions at the population scale²¹. Even though promising, ML-based methods used rarely to investigate maternal health services issues in Burkina Faso, especially when analyzing SBA²². Moreover, ML-based spatial prediction analysis and rural–urban subgroup analysis is hardly used in Burkina Faso¹⁹. The importance of determining geographic hotspots of low SBA coverage and the differences in predictors in rural and urban populations can be paramount to ensuring targeted interventions and allocating resources efficiently^{23,24}. The inclusion of such analyses could help to provide an evidence-based policy and programmatic planning that offers a spatially and contextually sensitive understanding of maternal health behavior²⁵.

Specifically, the objectives are to:

1. Identify and rank key associated predictors of SBA using supervised machine learning models;
2. Assess provinces-wise spatial inequalities in SBA coverage through prediction-based mapping; and
3. Examine rural–urban inequalities of SBA to inform context-specific maternal health interventions.

By integrating machine learning, spatial analysis, and subgroup exploration, this study aims to provide actionable insights into behavioral predictors and inequities in SBA utilization, supporting data-driven strategies to improve maternal healthcare coverage in Burkina Faso.

Methodology

Study design, data sources, setting, and participants

This cross-sectional study analyzed nationally representative secondary data from the 2021 Burkina Faso Demographic and Health Survey (BF-DHS) and accompanying geospatial data. The BF-DHS employed a stratified two-stage cluster sampling design to ensure representativeness at national, regional, and urban–rural levels. In the first stage, enumeration areas (EAs) were selected using probability proportional to size, with stratification by region and urban–rural classification. In the second stage, households were systematically sampled within each EA, yielding a nationally representative household sample²⁶. No patients are included in this study.

Study Sample

The analysis was restricted to women aged 15–49 years who had at least one live birth in the five years preceding the survey and whose most recent child was alive, with complete information available for the outcome variable and study predictors. The final sample comprised 5,111 ever-married women [Fig. 1].

Outcome variable

The primary outcome of this study SBA was defined as delivery accompanied by a trained and accredited health provider, such as a doctor, nurse, midwife, or auxiliary health staff, according to WHO and DHS criteria. Delivery accompanied by a traditional birth attendant, a community health volunteer, a relative, a friend, or without anyone present was considered unskilled birth attendance. The variable was considered a binary variable, where skilled birth attendance referred to receiving professional delivery service, while unskilled birth attendance referred to a lack of professional delivery service²⁷.

Conceptual framework and interpretation

It is important to clarify that this study adopts a predictive modeling framework rather than a causal inference approach. The machine learning algorithms employed are designed to identify patterns and associations that enhance the prediction of SBA. Accordingly, the variables identified in this study should be interpreted as predictive factors associated with the outcome, rather than causal determinants. Establishing causality would require alternative methodological approaches, such as experimental or quasi-experimental designs, which are beyond the scope of this study.

Machine learning framework and analytical strategy

Input features

The independent variables were categorized based on the existing literature, DHS recoding strategies, and meaningful groups to ensure analytical relevance and interpretability. Continuous variables, such as age, household size, maternal age at first birth, and age at first sexual intercourse, were coded into standard categorical groups,

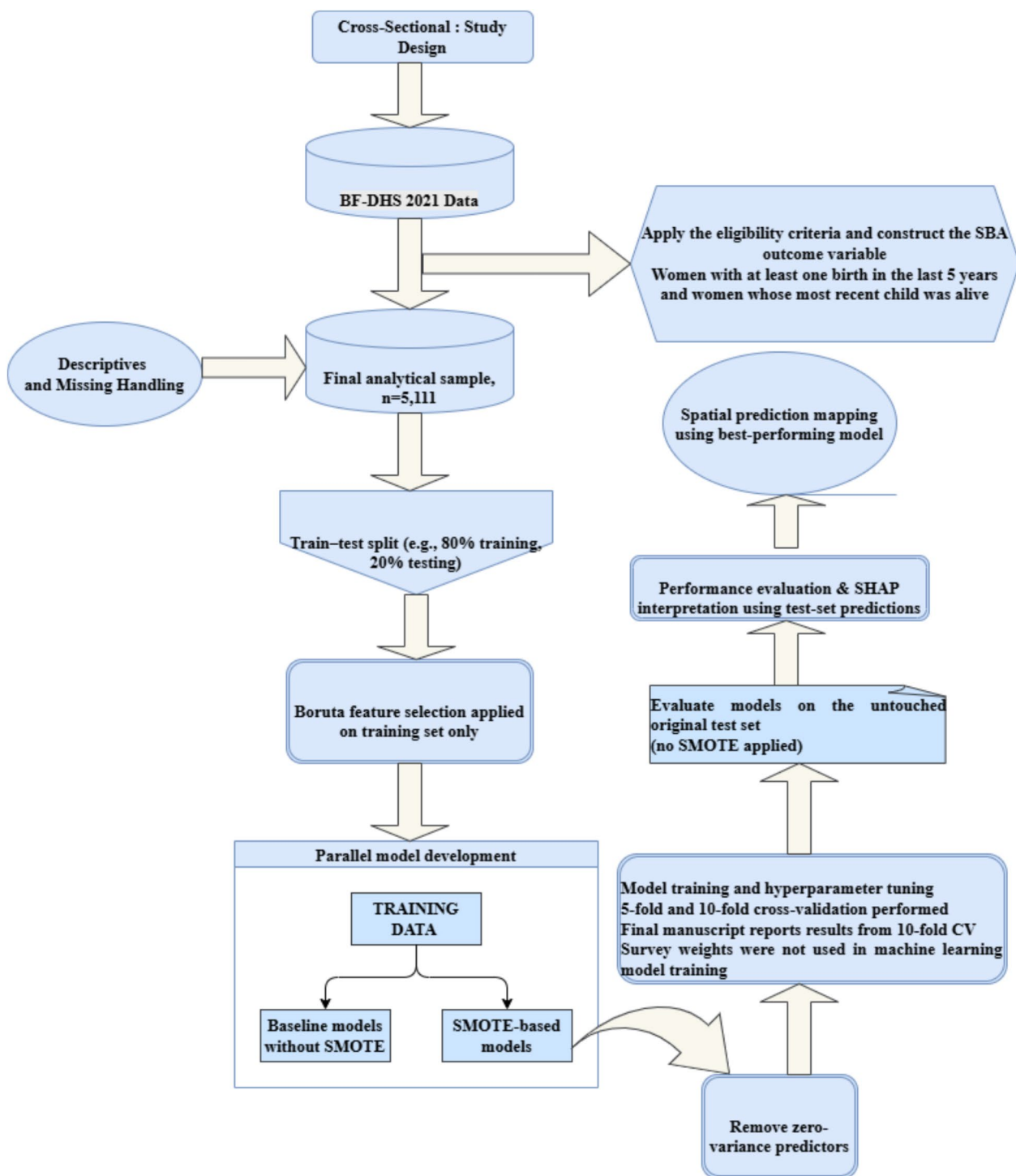


Fig. 1 Study flow.

while socioeconomic, reproductive, healthcare utilization, and empowerment-related variables were recoded into clinically and policy-relevant categories^{28–32}.

Features selection

Boruta algorithm was only used for feature selection to find all-relevant features, while RF was further evaluated as an isolated predictive model among other machine learning models with several learning perspective. Boruta feature selection was applied exclusively on the training dataset after data partitioning to prevent information leakage. Boruta is a robust wrapper-based algorithm built on random forests that evaluates each predictor against randomly permuted “shadow” variables³³.

Missing data handling

Missing data were low to moderate, with the highest missingness observed for husband’s education (8.9%), followed by pregnancy-related decision-making and pregnancy pressure (~5%), while all other variables had <1% missingness. Missing values were handled using multiple imputation by chained equations (MICE) under a missing-at-random assumption, with imputation models tailored to variable type. For summarizing imputation diagnostics and descriptive analyses Rubin’s rules were applied where appropriate³⁴.

Data partitioning and leakage prevention

The analytic dataset was split into training (80%) and test (20%) sets using stratified sampling to preserve the original class distribution of SBA. This separation ensured that all model development steps including preprocessing, resampling, and tuning were conducted solely on the training data, while the test set was reserved for unbiased performance evaluation.

Survey design considerations

The DHS uses a complex survey design with stratification, clustering, and sampling weights to ensure national representativeness. In this study, sampling weights were applied in descriptive analyses. All machine learning models, including cross-validation, were conducted on unweighted data, as is common in predictive modeling focused on classification performance rather than population inference. The DHS cluster structure was not explicitly modeled in the algorithms, although it is preserved in the original sampling design²⁴. The use of survey weights in machine learning remains methodologically challenging and lacks standardized implementation. This is acknowledged as a limitation, as the absence of weights and explicit cluster adjustment may introduce bias in population-level interpretation.

Data preprocessing

A recipe-based preprocessing pipeline was implemented to ensure reproducibility and consistency across all models. Categorical predictors were converted into numeric dummy variables using one-hot encoding. Predictors exhibiting zero variance were removed, as they do not contribute to model learning and may introduce numerical instability. Given the imbalance in SBA outcomes, the Synthetic Minority Over-sampling Technique (SMOTE) was applied only to the training dataset to address class imbalance without contaminating the test data. Class distributions and missing values were re-evaluated after SMOTE to confirm data integrity³⁵. SMOTE was preferred over class-weighted approaches due to its ability to generate synthetic minority samples, improving classifier learning across all models in a consistent manner. Additionally, model performance was evaluated on the original

imbalanced dataset without SMOTE to provide a baseline comparison, as reported in the supplementary materials [Supplementary Table S3].

Collinearity

Cramér's V correlation matrix was used to assess pairwise association between categorical predictors. Low to moderate inter-variable correlations were found in the analysis, indicating that there was no significant multicollinearity among the predictors [Supplementary Figure S7].

Model development

This study used five machine learning algorithms to compare and train: 1. RF—to establish nonlinear relationships and high-order interactions²¹. 2. DT (rpart) to obtain an interpretable baseline model³⁶. 3. KNN- to capture local patterns based on similarities³⁷. 4. LR to obtain a conventional benchmark model³⁸. 5. SVM to approximate irregular decision boundaries³⁹. Training of all models was done by use of k-fold cross-validation on the training set (5 and 10 folds), and 10-fold report in manuscript to ensure robustness and reduce overfitting. The grid-based search strategy was used and hyperparameters searched (tuneLength=10) to optimize the performance of the model and reduce the issue of overfitting in training dataset only⁴⁰. [see in Supplementary Table S2].

Classification performance metrics

Multipolar metrics were used to assess model performance to obtain an overall measure of classification quality, error trade-offs, resilience to class imbalance and discriminative capabilities. The accuracy of observations is the total percentage of correctly classified observations and gives a general idea of the performance in classification⁴¹. Precision alone, however, can be false in skewed data⁴². Precision is a measure of the ratio of the number of cases that are predicted to be positive to the actual positive cases. It is applied to evaluate the level of false positive errors. The recall (sensitivity) measures how effectively the model will identify true positive cases correctly and shows how many cases are falsely negative⁴³. Accurately, precision and recall describe the trade-off between falsely classifying negative cases as positive and not classifying positive cases. The F1-score is the harmonic mean between precision and recall, which is a balanced summary measure where the distributions of classes are unequal, and both the false positives and false negatives are significant⁴⁴. The MCC is a more detailed metric of binary classification performance since it uses each of the four elements of the confusion matrix. The Kappa of Cohen is the measure of the relationship between the observed and the predicted classifications of the samples larger than what is predicted by mere chance⁴⁵. Quantitative calibration performance was additionally evaluated using brier scores. This measure will be useful in assessing the reliability of the model, particularly when the prevalence of classes is not equal. Lastly, AUROC was used to measure discriminative ability to differentiate between positive and negative results at all conceivable classification thresholds regardless of the distribution of classes⁴⁶.

Discrimination, calibration and clinical utility

AUROC curves were used to assess model discrimination and sensitivity–specificity trade-offs across models. Calibration was evaluated using calibration plots by associating observed SBA proportions with predicted probabilities from each model in the test dataset. Therefore, raw predicted probabilities were used without post-hoc calibration approaches such as Platt scaling or isotonic regression. Model calibration was measured using the Brier Score, with values closer to 0 indicating better-calibrated predictions. To assess the potential public health utilization of the best performing model, this study used a DCA using the *rmda* package in R studio based on the RF, LR, and SVM models were assessed on the testing dataset. The DCA plots the decision boundary between

probability ranging 0 to 1 and can be compared with either the treat all or treat none strategy. In this framework, DCA was conducted to assess the potential effectiveness of the models in identifying women at higher risk of unskilled birth attendance who may benefit from targeted maternal health interventions. For policymakers or maternal health planners considering distribution of complementary maternal health assistance based on predicted risk thresholds using decision framework indicates programmatic decision-making⁴⁷.

Model interpretability

The most successful model was presented with SHAP in order to be more transparent and able to interpret the policy. SHAP analyses included mean absolute SHAP values as a global feature importance measure. Local explanations, which showed how individual predictors affected particular predictions. Cumulative SHAP contribution plots, determining the subgroup of predictors with most effective explanation of model variance. This interpretability system allowed to achieve the definition of the main behavioral, socioeconomic, and healthcare-related determinants of skilled birth attendance⁴⁵.

Spatial prediction and subgroup analysis

The SBA predicted probabilities from the best-performing model were estimated at the individual level and subsequently aggregated at the provincial level to generate spatial prediction maps, enabling visualization of geographic inequalities and areas of comparatively low SBA coverage. From the better performing model, the spatial model was estimated subsequently, combined at the province level to provide spatial prediction maps, indicating inequalities of rural and urban areas. Provincial estimates were measured as the mean predicted probability within each province. Therefore, a separate model was conducted for rural and urban residents. No additional modeling was used in this study. Additionally, no uncertainty, inferential spatial modeling, or spatial smoothing were used; thus, the findings maps were constructed as descriptive visualization rather than spatial inference.

Sensitivity analysis

Additionally, a survey-weighted logistic regression models were fitted in STATA-17 integrating sampling weights (v005), primary sampling units (v021), and stratification (v022) employing the svyset command for performing sensitivity analysis. This additional method used for the complex DHS survey design and provides evaluation of the robustness of the results to survey weighting; The findings of this analysis are accessible in the Supplementary Materials and were constant with the main machine learning findings [Supplementary Table S4].

A sensitivity analysis applied to assess the validity of the key RF model and evaluate the role of different class imbalance handling approaches on predictive performance and feature importance.

Class imbalance

The training dataset showed notable class imbalance, with the minority ("Unskilled") class accounting for 9.6% of observations and the majority ("Skilled") class comprising 90.4%. All models were implemented using the Boruta-selected input features.

SMOTE-based random forest

The primary RF model conducted the SMOTE to stability the training data at a 1:1 ratio. The model was trained using 500 trees with the default *mtry* parameter and assessed on an independent test set¹⁰.

Class-weighted random forest

A cost-sensitive RF model was implemented by transmission reverse class-proportion weights to penalize misclassification of the minority class. The model was trained using the same hyperparameters as the primary RF model³⁵.

Balanced random forest

A balanced RF model was applied using the *ranger* package. Balanced bootstrap sampling was used to confirm equivalent class representation in each tree, with 500 trees and permutation-based variable rank estimation²⁴.

Analytical tools

All analyses were performed in R (version 4.5.5). The randomForest package was used for the SMOTE-based and class-weighted RF models, whereas the ranger package was used for the Balanced Random Forest. Model performance was evaluated using the caret and pROC packages.

Code availability

The analytical scripts and reproducible workflow are publicly available through GitHub and archived in Zenodo (<https://doi.org/10.5281/zenodo.22712479>). Source code: <https://github.com/muhammadsalek/SBA-BurkinaFaso-ML-DHS>.

Results

Background characteristics

Only 9.6% delivered without skilled assistance [Supplementary Figure S1]. Most of the women had skilled birth attendance, were aged between 20–34 years (71%), from rural areas (69%), and had no formal education (64%). Most of the women were multiparous (78%), had ≥ 4 ANC visits (72%), delivered vaginally (93%), and reported relatively low health insurance coverage of 0.55%. [Supplementary Table S1]. The original data has 90.4% skilled vs 9.6% unskilled, while the synthetic training set created by SMOTE, on the left, has 50% skilled and 50% unskilled. The SMOTE remedies class imbalance by oversampling synthetic "unskilled" cases, thus improving its detection capability of high-risk women without biasing the evaluation that needs to be performed on the original test set [Supplementary Figure S2].

Results of feature selection

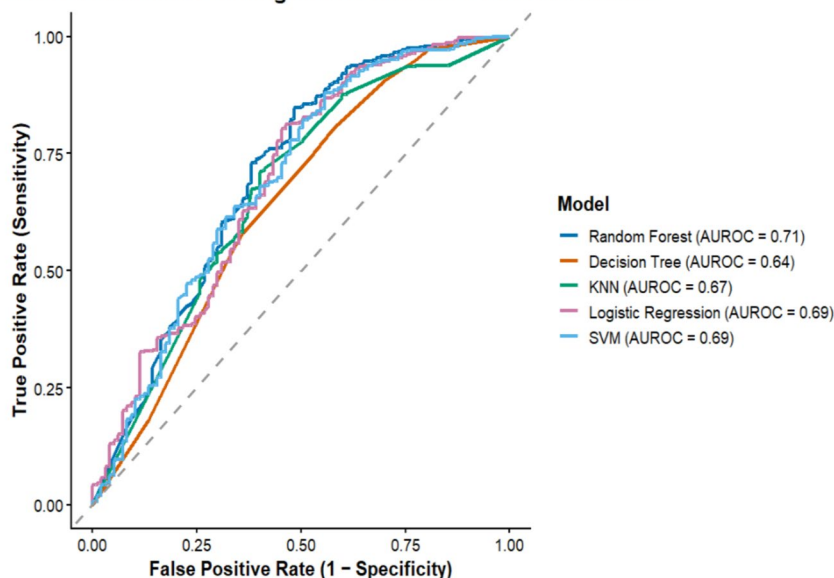
In the analysis of SBA, the Boruta feature selection algorithm confirmed 17 predictors as statistically significant and relevant for SBA such as province, ethnicity, wealth category, antenatal care visits, working status, religion, media exposure, internet use, maternal age at first birth, place of residence, age at first sexual intercourse, parity, current sexual activity, child-related category, household size group, fertility preference, and women's decision-making autonomy. [Supplementary Figure S3, Supplementary Figure S10].

By the end of the hyperparameter tuning using 10-fold cross-validation, the RF model showed ranked among the best and achieved the maximum AUROC=0.71, F1-score=0.94, MCC=0.31, and Kappa=0.30. Due to high class imbalance, accuracy may be misleading; thus MCC, Kappa, AUROC, and PR-AUC were used as main

Table 1 Performance metrics of machine learning models for predicting SBA.

Model	Accuracy	Precision	Recall	F1	MCC	Kappa	AUROC
Random forest	0.89	0.93	0.96	0.94	0.31	0.30	0.71
Decision tree	0.85	0.92	0.90	0.91	0.19	0.19	0.64
KNN	0.76	0.94	0.79	0.86	0.19	0.17	0.67
Logistic regression	0.71	0.94	0.72	0.82	0.19	0.15	0.69
SVM	0.88	0.93	0.94	0.93	0.27	0.27	0.69

Fig. 2 Comparison of AUROC.

ROC Curves of Machine Learning Models for Skilled Birth Attendance


assessment metrics. The competitive behavior of performance depicted by SVM could not outperform the RF model, as it provided a more balanced and robust classification performance in all evaluation metrics [Table 1].

The baseline model performance without class imbalance correction, showing generally weak discriminatory performance across all models (AUROC: 0.51–0.59; MCC: 0.05–0.10) [Supplementary Table S3].

Model discrimination and comparative performance

The ROC analysis showed that the RF model showed the moderate discriminatory power in predicting SBA, with an AUROC of 0.71, moderate the DT with 0.64, the KNN with 0.67, the LR with 0.69, and the SVM equally with 0.69. An AUROC=0.71 indicates moderate discrimination, suggesting the RF model assigns a higher predicted probability of SBA in approximately 71% of randomly selected pairs comprising one woman who received SBA and one who not [Fig. 2].

Sensitivity analysis results

The sensitivity analysis performed on three methods to deal with imbalance indicated significant differences in model performance (Supplementary Table S5). The RF revealed through the application of SMOTE provided the best AUROC (0.716), sensitivity for minority class (96.1%) and overall accuracy (89.5%), and had the smallest Brier score (0.091). The lowest performance was demonstrated by the Class-weighted RF (AUROC: 0.66; Sensitivity: 68.4%) [Supplementary Table S5, Supplementary Figure S8, S9, S11].

Calibration analysis showed that the RF and SVM models produced well-calibrated probability estimates, with a very close approximation to the ideal calibration line, while DT and KNN models slightly overestimated at their lower predicted probabilities [Fig. 3].

Across all models, precision remains relatively stable at moderate recall levels, with a decline at higher recall, reflecting the inherent trade-off between sensitivity and precision in highly imbalanced classification settings (Supplementary Figure S5).

Overview of global feature importance—SHAP summary plot

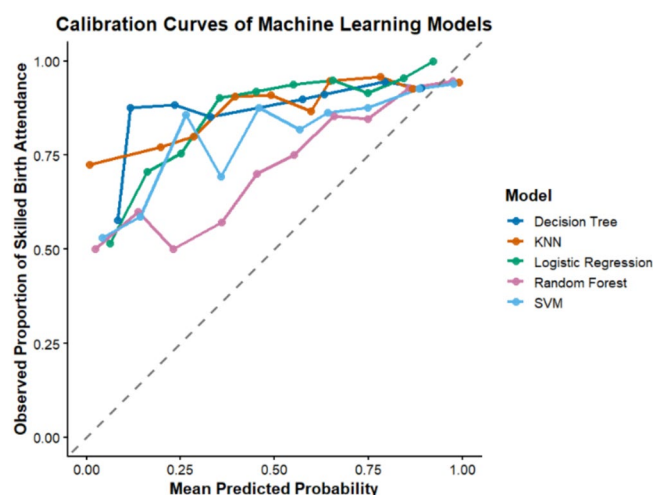
A small subset of sample Sahel province, antenatal care utilization ≥ 4 visits, maternal age at first birth ≥ 20 years, current sexual activity, and age at first sexual intercourse ≥ 18 years contributed most strongly to models' predictions. These findings reveal the multidimensional nature of skilled birth attendance, indicating geographical location and socioeconomic status, with behavioral, reproductive, and information-related factors. Additionally, the most influential predictor in the SHAP analysis was province. Women living in the Sahel revealed lower predicted probabilities of SBA, while Centre-Est and Sud-Ouest indicated higher predicted probabilities, suggesting notable geographic disparities across administrative provinces [Fig. 4].

Model parsimony and classification performance

Cumulative SHAP contribution

By using the mean absolute SHAP values averaged across all observations the predictor importance was evaluated. Additionally, the cumulative contribution percentages were measured by normalizing the totaled mean about SHAP values across predictors relative to the total SHAP magnitude. Predictor rank was evaluated using the mean absolute SHAP values averaged across all observations. Cumulative contribution percentages were calculated by normalizing the totaled mean absolute SHAP values across predictors relative to the total SHAP magnitude. The top 8–10 predictors accounted for approximately 50% of the cumulative normalized SHAP contribution, such as geographic setting (particularly Sahel province), antenatal care utilization (≥ 4 visits), maternal age at first birth ≥ 20 years, sexual activity status, age at first sexual intercourse, household wealth, and household religion. These results highlight that a relatively small subset of predictors dominated the model decision-making process. As additional predictors were included, cumulative contribution increased more gradually, with approximately

Fig. 3 Calibration plot.



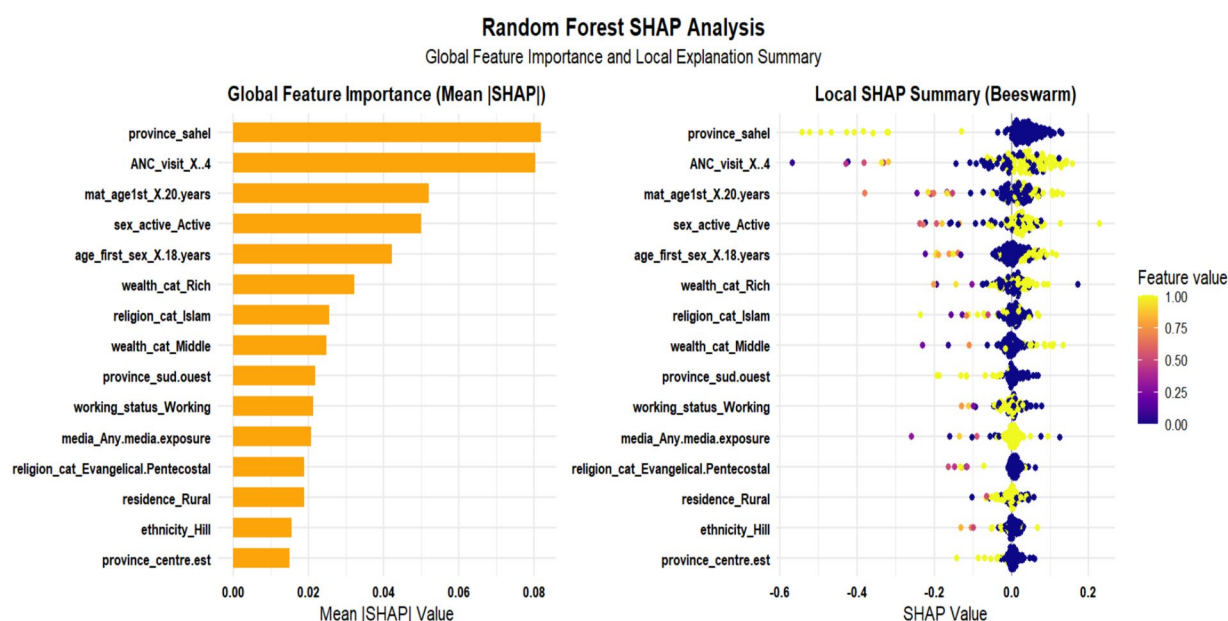


Fig. 4 Feature importance based on mean absolute SHAP values and Local explanation summary.

20–25 variables accounting for nearly 80–90% of the total normalized SHAP contribution [Supplementary Figure S4]. Beyond this point, additional predictors contributed only marginal incremental explanatory value.

Confusion matrix of the random Forest

For the women who actually delivered with a skilled birth attendant, 448 were correctly classified as being skilled, while 95 were misclassified as being unskilled. Of the women who delivered without skilled assistance, 188 were correctly classified as being unskilled and 49 were falsely classified as being skilled. The distribution indicates a high sensitivity that reflects the model's ability to correctly identify most of the women who obtained skilled delivery care, along with more moderate specificity [Supplementary Figure S6].

Clinical utility and spatial distribution of predicted SBA

Decision curve analysis (DCA)

The model's net benefit curve consistently lay above both the "treat-all" and "treat-none" reference strategies for threshold probabilities approximately between 0.10 and 0.80, indicating a positive standardized net benefit throughout this range. In practice, applying the model to guide intervention targeting would result in a higher proportion of cases identified and intervened upon (while avoiding unnecessary interventions) than non-selective approaches across a wide spectrum of choice for risk–benefit trade-offs. Therefore, the model can be used to support risk-stratified strategies, such as prioritizing women for community outreach, referral support, transport vouchers, or enhanced antenatal follow-up, without imposing an excessive burden on low-risk women [Fig. 5].

Spatial distribution of predicted SBA: Nationwide patterns

Higher predicted probabilities were exhibited in Centre, Centre-Nord, Centre-Ouest, Centre-Sud, Nord, Boucle-du-Mouhoun, and Cascades (approximately 0.60–0.80), while Sahel, Sud-Ouest, and parts of the eastern and

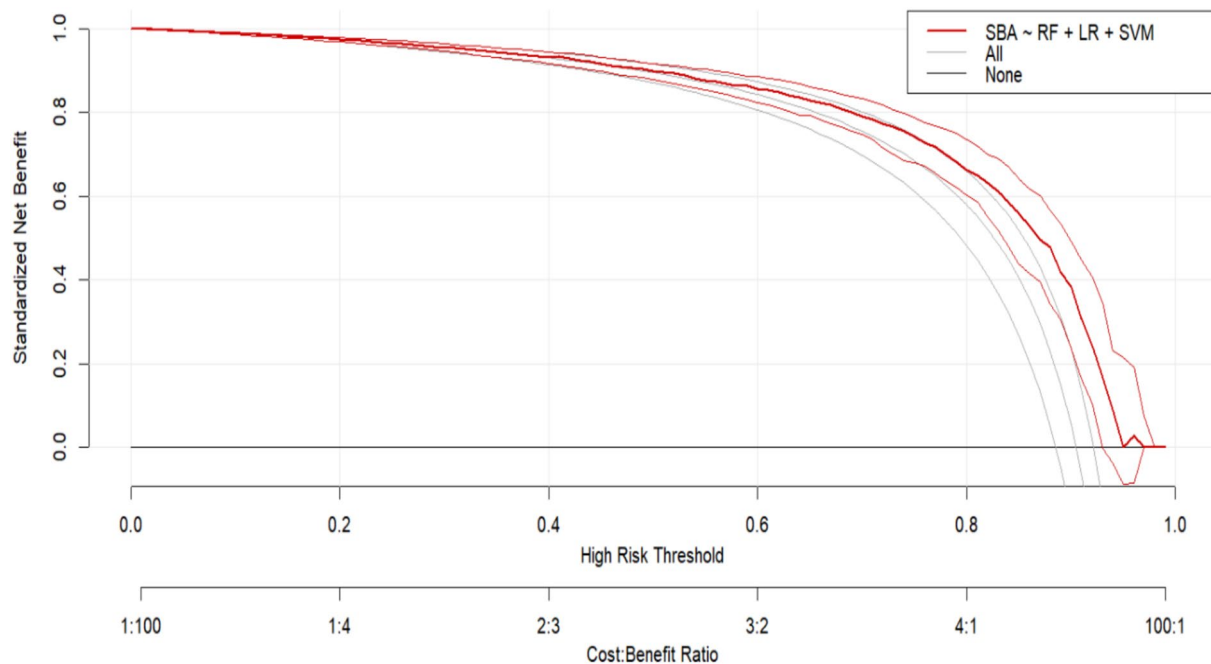


Fig. 5 Decision curve analysis comparing the net benefit of five machine learning models for predicting skilled birth attendance.

southwestern regions revealed lower predicted probabilities (approximately 0.20–0.40), indicating notable geographic disparities in SBA coverage [Fig. 6].

Urban–Rural stratified spatial patterns

Urban areas

Generally, the predicted probabilities of SBA higher in urban areas of Centre, Centre-Ouest, Centre-Sud, Centre-Nord, Hauts-Bassins, Boucle-du-Mouhoun, and Cascades, indicating improved access to health facilities. In contrast, urban areas in Sahel and Sud-Ouest observed lower predicted probabilities, indicating persistent structural disparities.

Rural areas

Rural areas revealed higher geographic inequalities as the Sahel, Est, Sud-Ouest, and parts of Centre-Est had the lowest predicted probabilities [Fig. 7].

Discussion

This study applied an interpretable machine learning framework to examine the key predictors and provinces-wise, and rural–urban wise descriptive spatial inequalities of SBA in Burkina Faso. The baseline results without SMOTE indicate poor minority class discrimination, as reflected by low MCC and near-random AUROC values, highlighting the importance of addressing class imbalance in predictive modeling.

Higher prevalence SBA in the analytic sample than in national estimates from DHS is likely explained by the narrow inclusion criteria and outcome construction for this study, which relied on the most recent live birth with

Fig. 6 Provinces-wise SBA inequalities. Maps were generated by the authors using R version 4.5.1 with sf and ggplot2. Administrative boundary shapefiles were obtained from the Humanitarian Data Exchange (HDX) Common Operational Datasets (COD) for Burkina Faso (<https://data.humdata.org/dataset/cod-ab-bfa>).

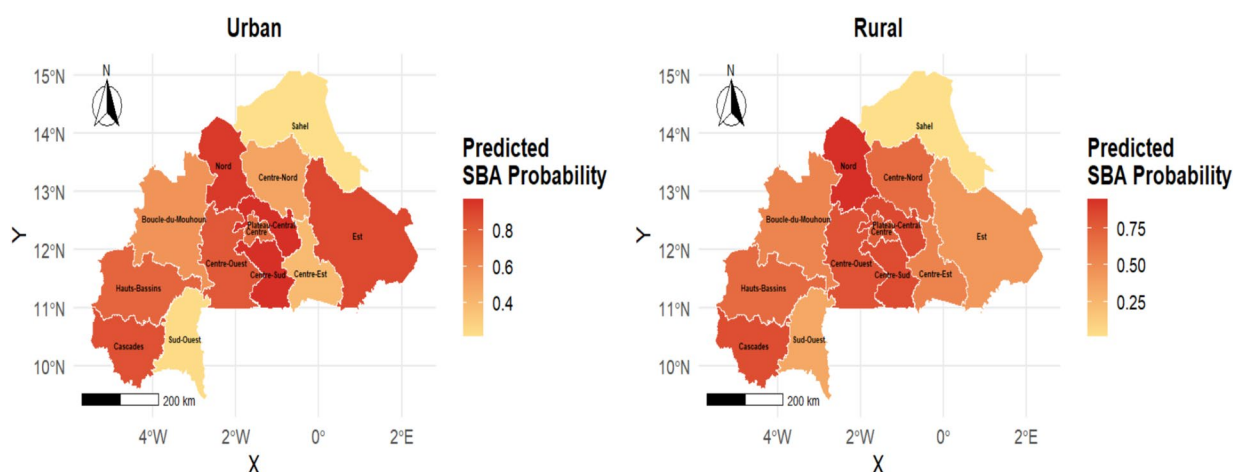
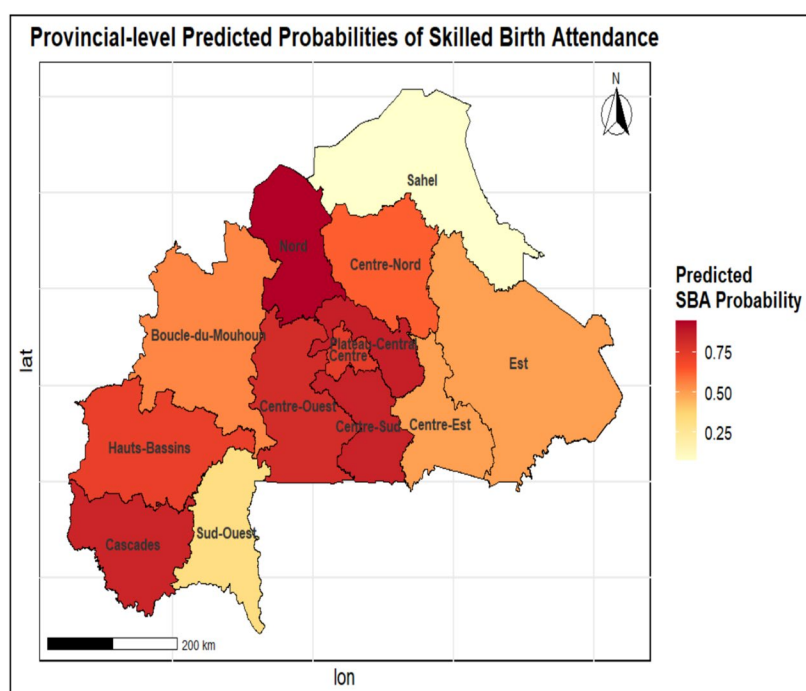


Fig. 7 Rural–urban wise spatial inequalities of SBA. Maps were generated by the authors using R version 4.5.1 with sf and ggplot2. Administrative boundary shapefiles were obtained from the Humanitarian Data Exchange (HDX) Common Operational Datasets (COD) for Burkina Faso (<https://data.humdata.org/dataset/cod-ab-bfa>).

complete data. They also recommend caution in interpreting the reported prevalence as a population-level prevalence value, but rather as an analytic subset of the population for predictive modeling. The accuracy was high, but the low MCC values and Kappa values showed that there was a high deal of class imbalance, and therefore, accuracy alone is misleading. Thus, the performance of the models must be evaluated mainly based on MCC, AUROC and F1-score which are more stable measures to evaluate performance under imbalanced classification scenarios²⁴.

Sensitivity analyses revealed coherent predictive performance across SMOTE-based, class-weighted, and Balanced RF models, supportive the validity of the primary findings. This coherent reveals that the selection of class imbalance handling approach had minimal influence on model discrimination and predictor importance.

The RF model demonstrated acceptable discriminatory performance and identified key predictors such as SBA utilization, including geographic location, antenatal care (ANC) utilization, maternal age at first birth, age at first sexual intercourse, parity, household wealth, and women's decision-making autonomy was associated with SBA. Administrative province, particularly residence the Sahel province showed the strongest factor predictor of RF model, suggesting that regional disparities in health care services access and maternal health systems associated with SBA. Conversely, provinces including Centre, Centre-Ouest and Sud-Ouest showed positive SHAP, indicating relatively predicted probabilities of SBA. These differences may associate with contextual variation across administrative province, including health care accessibility and regional health system characteristics. The use of ANC (4 or more) was also another predictive variable, which supports the study results that continued birth preparedness, health education and planning of facility delivery are promoted by regular antenatal contact. The identified low-performing provinces may benefit from targeted maternal health interventions, including strengthened ANC, skilled birth attendance outreach, and maternity waiting homes. Such place-based strategies may improve access to SBA services in underserved areas.

SBA was reported to be lower with early maternal age at first birth (<20 years) and early sexual initiation (<18 years), which indicates the risk of adolescent mothers due to the social, financial and cultural constraints they might have as a result of not receiving skilled assistance. Rich and employed women are better positioned to afford transport, informal costs, and time away from household responsibilities⁴⁸. Media and internet exposure likely enhance knowledge of maternal health benefits and normalize facility delivery, shifting norms away from traditional practices⁴⁹.

The estimated socioeconomic and geo-spatial inequalities in SBA utilization may be interpreted through Andersen's Health Utilization Model, where enabling factors such as household wealth, maternal education, and healthcare accessibility significantly shape maternal healthcare utilization behaviors⁵⁰. Additionally, the spatial inequalities observed in underserved regions may reflect the Three Delays Model, mostly delays in attainment healthcare facilities and receiving adequate obstetric care⁵¹.

The association between behavioral and structural determinants of maternal health service use is further supported by socioeconomic factors, such as wealth status, work status, exposure to the media, and so forth⁵². Women with limited ANC contact may lack both information and logistical support, associated with home delivery or reliance on unskilled attendants, particularly in rural settings⁵³. Younger mothers may also underestimate delivery risks or face restrictions on independent healthcare decision-making, increasing vulnerability to unskilled delivery⁵⁴. Women who participate in household decisions may be better able to seek skilled care, negotiate transport, or insist on facility delivery, particularly when labor begins unexpectedly⁵⁵.

Although there was a general high predicted probability in urban centers, still there were low coverage pockets even in urban centers, showing urban residence is not a solution to inequities. Small towns were even more contrasting with the poorest SBA probabilities in the peripheral provinces (Sahel, Est, and Sud-Ouest). Such disparities in the rural areas are probably due to complex geographic, socioeconomic, and systemic disadvantages, such as transportation difficulties, poverty, and poor referral networks⁵⁶. The observed geographic disparities, particularly in rural and peripheral provinces, highlight persistent inequalities in maternal healthcare access. These findings may help policymakers prioritize geographically targeted interventions and allocate resources to underserved regions.

This suggests that place-based disadvantage persists even within urban settings, where informal settlements, poverty, and overstretched facilities may limit access to skilled care⁵⁷. This kind of evidence shows the necessity of more geographically specific interventions as opposed to a national approach that fits all⁵⁸. Although these variables improved prediction accuracy, model outputs are only meant to provide population-level public health insights; they should not be construed as causal or discriminatory determinants.

The use of SHAP enhances model transparency by explaining how individual predictors contribute to SBA predictions rather than simply identifying high-risk women. This may support policymakers in designing targeted maternal health interventions based on the factors underlying predicted risk.

Policy implications and conclusions

High-burden areas, especially rural Sahel, should receive policy interventions that can be achieved through improving access to facilities, mobilization of skilled health workers and reinforcement of ANC initiatives. This study offers evidence-based recommendations on how inequities in SBA can be reduced and maternal health coverage can be improved in Burkina Faso using machine learning, spatial mapping, and subgroup analysis as a contribution to the national and global strategies of providing women with safe childbirth and safety in childbirth.

Limitations

There are some limitations in the current analysis. First, it is based on cross-sectional data, which is not adequate for ascertaining causality between predictors and skilled birth attendance (SBA). Additionally, even though SMOTE worked well as a handle class imbalance technique, it might not be representative of unskilled births as observed in actual settings. The analytical sample was restricted to women with complete information on the most recent live birth and study variables, which may lead to differences in SBA prevalence compared to national DHS estimates. Thus, the reported prevalence should not be interpreted as a population-level estimate, and some degree of selection bias may be present. A key limitation is that the machine learning models were developed using unweighted data, which may limit population representativeness and potentially introduce bias in estimated predictor importance. Future studies may additionally compare penalized regression and gradient boosting methods such as LASSO or XGBoost. confidence intervals (CI) for model performance measures were not computed since the experiment involved only one splitting into training and testing sets. Thus, any possible estimation of the uncertainties of the model performance was not quantified, and further studies might include bootstrapping techniques.

Strengths

In spite of these challenges, there are a number of advantages in this study. The research uses a large and representative sample ($N=5,111$), so findings would be generalizable across Burkina Faso. Applying meaningful interpretable machine learning (Random Forest with SHAP values) facilitates a clear and thorough determination of important factors and is a significant advantage over regression analysis. The blending of spatial analysis provides clear and meaningful information regarding geographical inequality and would highlight important aspects of inequality in urban and rural areas.

Future work recommendations

Future studies could include data related to facility, healthcare system, and geographic accessibility, which could help in improving prediction capability and analysis related to the underlying structures. Longitudinal studies could shed light on the causal relationship of maternal health practices with behaviors related to service use. Qualitative studies could help in understanding underlying factors related to SBA in high-risk provinces.

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Author contributions Md Salek Miah conceived and designed the study, curated and managed the data, performed all statistical and machine learning analyses, interpreted the findings, and drafted and critically revised the manuscript.

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Data availability All the data sets used in this research can be accessed free at the Burkina Faso DHS website, [<https://dhspogram.com>]. The code for analysis and SHAP analysis visualization scripts can be obtained from the corresponding author on reasonable request.

Declarations

Competing interests The authors have no competing interests to declare.

Ethics In this study, data collected from the Demographic and Health Survey program, which is an anonymized survey data set available to the public, was employed. The use of the data did not require additional ethical clearance, as the data is anonymized and the clearance and consent to participate. Ethical approval for the 2021 BF-DHS was obtained from national ethics committees in Burkina Faso and the ICF International institutional review board. No additional approval was required for this secondary data analysis. However, details available at : <https://dhsprogram.com/Methodology/Protecting-the-Privacy-of-DHS-Survey-Respondents.cfm>.

Transparency statement The lead author declares that this manuscript is a truthful and transparent report of the study being described. There have been no crucial events or aspects of the analysis missing from this manuscript, nor have any deviations from the analysis not been reported.

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